

Marine microbiology

Winter Semester

Lecture 5

Advantages and Disadvantages of marine microorganisms

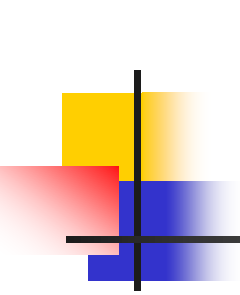
References:

Prescott's Microbiology (PM): Chapter 35

Munn. Marine Microbiology (MM): Chapter 11-14

第八章 MOOC（四节内容）

第九章 MOOC（三节内容）



Marine microbes have some detrimental (bad) effects:

- **Diseases (as pathogens)**
- **Biofouling** 生物污损
- **Biodeterioration** 生物腐蚀

What can we do about them? Potentially good approaches are:

- **Bioremediation** 生物修复
- **Biotechnology** 生物技术

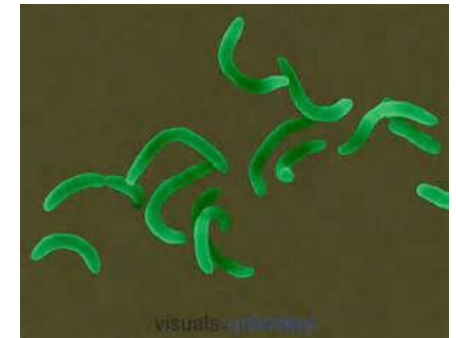
1. Diseases of human

Vibrio species are major human pathogens

- The vast majority of marine microbes are non-pathogenic for humans
- The most important human diseases associated with marine bacteria are caused by members of genus ***Vibrio*** 弧菌
- Vibrios are Gram negative, curved rods, γ -proteobacteria
- About 12 species of *Vibrio* have been associated with human disease.

The most important are:

- *Vibrio cholerae* 霍乱弧菌
- *V. vulnificus* 创伤弧菌
- *V. parahaemolyticus* 副溶血性弧菌
- Other species are occasionally associated with diarrhoea or wound infections, usually following seafood contact
 - *V. harveyi* 哈氏弧菌
 - *V. alginolyticus* 溶藻弧菌



Scombroid_{鯖鱼} fish poisoning is a result of bacterial enzyme activity

- Associated with eating fish of family Scombridae: tuna (as in sushi), mackerel, swordfish
- Fish tissue contains high level amino acid **histidine** 组氨酸
- If storage temperature too high, bacteria in fish convert histidine to **histamine** 组胺
- Bacteria associated with this include *Photobacterium phosphoreum* 明亮发光杆菌
- Histamine induces **allergic response**
 - shortness of breath, respiratory failure may occur



Clostridium botulinum (肉毒杆菌) causes botulism: a rare lethal intoxication from seafood

Clinical symptoms and virulence factor

- **Intoxication** 中毒 = ingestion of microbial toxin causes the symptoms, microbe does not need to be present or colonize host
- *C. botulinum* produces a **neurotoxin: botulinum toxin**
 - the most lethal toxin known: 200 ng can be fatal
 - causes **paralysis** 麻痹
 - flaccid (floppy) paralysis – muscles cannot contract

Epidemiology

- Anaerobic spore-forming Gram positive bacteria, in marine sediments and intestinal contents of fish and crustaceans
- Most cases from home-canned foods
- Outbreaks in Arctic Inuit people from fermented whale and seal meat. Outbreaks from smoked fish (smoking kills most but not all spores) if not refrigerated



Toxic dinoflagellates and diatoms pose serious threats to human health

- Some planktonic microbes able to form blooms produce toxins
- Frequently referred to as “red tides”(赤潮), although not all toxic blooms are red
- Now termed “harmful algal blooms” or **HABS**
- **Only about 150** of the thousands of phytoplankton species are known to produce toxins
- Health hazard from eating fish or shellfish that have accumulated toxin in their tissues
- Some toxins produce disease symptoms as a result of direct contact with water
- **Toxins are not proteins, and remain active after cooking: most are neurotoxins**



The incidence of HABs and toxin-associated diseases is increasing

2. Diseases of marine mammals

a) **Dinoflagellate** and **diatom** toxins affect marine mammals, birds

甲藻

硅藻

- HAB toxins effects as for humans: mainly effects on nervous system
- One explanation for **stranding of whales and dolphins**
搁浅
 - disoriented due to accumulation of toxins after feeding on contaminated fish or shellfish
 - high levels **saxitoxins** found in stranded killer whale tissues
 - saxitoxins and **brevetoxins** bind to nerve tissue of marine mammals
- Toxin domoic acid produced by diatom ***Pseudo-nitzschia*** causes outbreaks of neuropathological illness in marine mammals and birds
 - Amnesia 健忘症 in humans
 - Mass mortalities of sea lions, pelicans USA
 - **Domoic acid** enters food chain, krill accumulate it, passed to higher levels

拟菱形藻

鹈鹕

磷虾



Pilot whale mass stranding



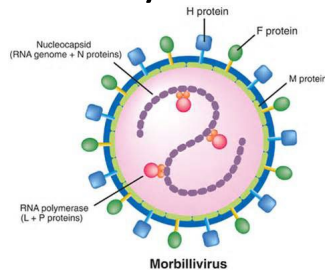
Pseudo-nitzschia

2. Diseases of marine mammals

b) Viruses from 9 families linked to diseases of marine mammals

■ **Morbilliviruses** 麻疹病毒

- ssRNA viruses, spherical enveloped; also cause measles 麻疹 (humans), distemper 犬瘟热 (dogs)
- Most important viruses infecting cetaceans (whales, dolphins) and pinnipeds (seals, sea-lions)
- Cause pneumonia 肺炎, disturbance of swimming and navigation, rapid death may occur



2. Diseases of marine mammals

c) Several species of bacteria and fungi infect marine mammals

- ***Leptospira*** 钩端螺旋体

- Causes leptospirosis – kidney failure in seals and sea-lions

- ***Mycobacterium***

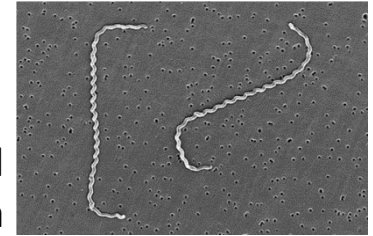
- *M. tuberculosis* and *M. bovis* cause tuberculosis 肺结核
- *M. marinum* causes lesions in skin and lungs
- Occurs in seals, sea-lions and cetaceans
- Transmission to humans can occur

- ***Aspergillus***

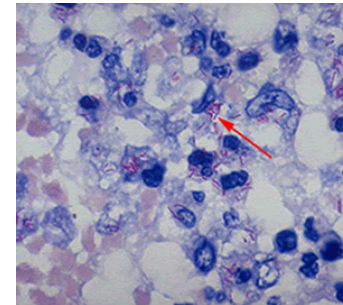
- Fungus can cause pneumonia 肺炎 in stressed captive marine animals

Aspergillus spores

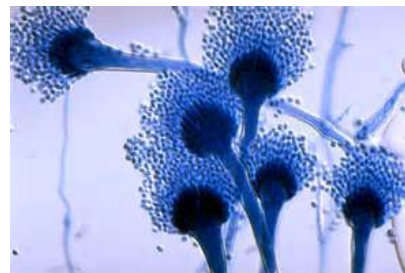
Spiral
bacteria



M. marinum skin lesions



Intracellular *Mycobacterium*



3. Diseases of Invertebrates

1) Diseases of corals



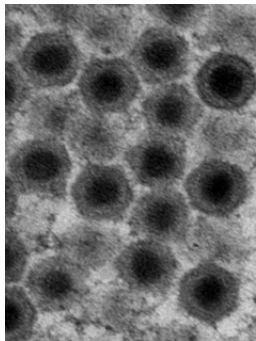
- Diseases have now been described in all the major reefs systems of the world
- Many coral researchers believe diseases of corals are result of stress effects on zooxanthellae (虫黄藻) e.g. high temperatures, pollution, bacteria. (see MM p. 230 for discussion)
- The causative agents or pathogens responsible for most of the coral diseases are not known
- Some biologists are sceptical about the role of microbes as primary causative agents of coral disease. A small number of diseases are well-characterised with strong evidence of microbial association

2) Diseases of Molluscs

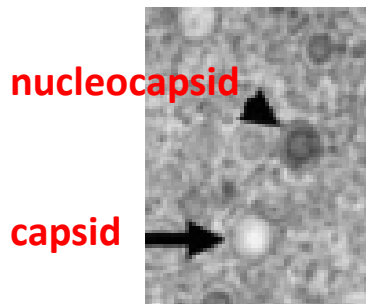
(软体动物)



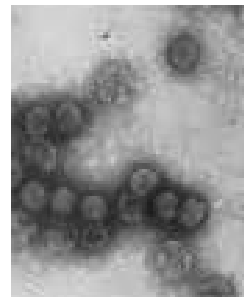
- a) **Vibrios** are a major cause of diseases in cultured molluscs
 - Virulence factors are *exotoxins*: hemolysins 溶血素, proteases
- b) A wide range of other bacteria can cause infections in bivalve molluscs
- c) **Viral infections** are a major problem in oyster culture



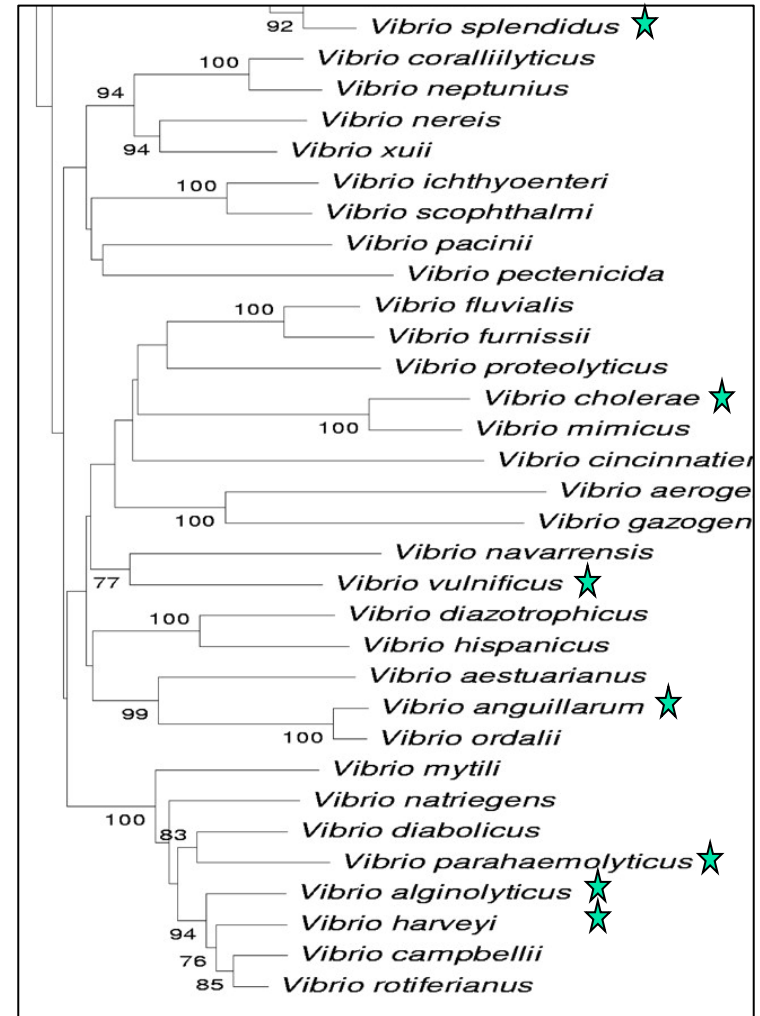
Iridovirus
虹彩病毒



Ostreid herpes-virus
OsHV-1



marine birnaviruses
MABVs



Part of a *Vibrio* phylogenetic tree

3) Diseases of Crustaceans

(甲壳类)



- Crustacean diseases are important in both wild and cultured populations of lobsters, crabs, shrimp
- Expansion of intensive prawn culture has been accompanied by a dramatic spread in viral diseases
 - well-characterised viral pathogens from 6 in 1980s to about **20 viruses** today
- Some crustacean species feed on each other or on feces, leading to rapid disease transmission
- Initial lack of government regulation led to severe problems of pollution and disease
- Nowadays the development of gene probes and PCR methods is assisting in control methods

3) Diseases of Crustaceans

Viral diseases:



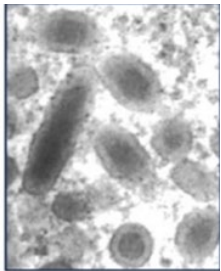
Whispovirus (白斑病毒)
in penaeid prawns.



Yellowhead disease virus (YHV)
in giant black tiger.



Taura syndrome virus (TSV)
(陶拉综合征病毒) in white shrimp.

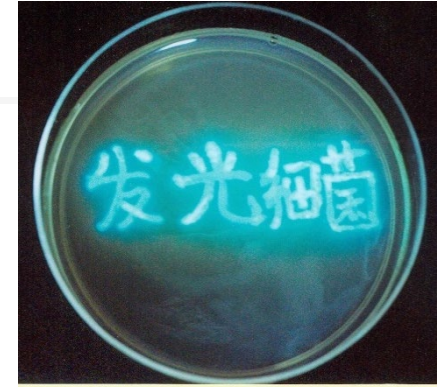


Presence in Asia-Pacific

3) Diseases of Crustaceans

Bacterial diseases of prawns:

- **Vibrios are the major causes** of devastating losses in hatcheries and ponds of shrimp and prawn aquaculture – world-wide
 - The main pathogen is *V. harveyi*; others include *V. vulnificus*, *V. parahaemolyticus*, *V. alginolyticus*
 - Because of its luminescence 发光, outbreaks termed “**luminous vibriosis**”
 - Lesions vary: black cuticle lesions, black lymphoid organ, red body and gills etc.
 - Mortality ranges from minor up to 100%



4. Diseases of Fish



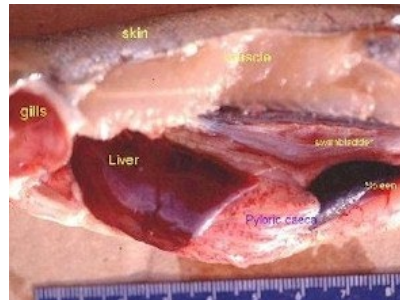
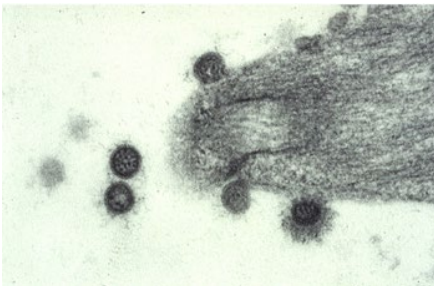
- Microbial diseases of fish cause major losses in aquaculture
- Effects on natural fish populations are harder to determine
- Large-scale fish kills in the wild occur
 - First scientific description was “red pest” in eels, Italy, 1718
 - Now known as vibriosis – *Vibrio anguillarum* 鳗弧菌
 - Caused mass mortalities in migrating eels



a) Viruses cause numerous diseases of marine fish

▪ Infectious salmon anemia virus (ISAV)

- One of the most important pathogens in salmon culture



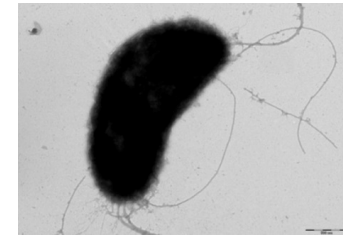
- Anemia 贫血 (pale gills),
- hemorrhages in eye
- Liver and spleen are swollen 肿胀 and dark

4. Diseases of Fish

b) Bacterial diseases of prawns

Vibrios are a major cause of diseases in marine fish

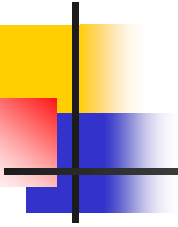
- ***Vibrio anguillarum*** 鳎弧菌
 - **The first vibrio identified as a fish pathogen, causes vibriosis in eels**



- ***Other vibrios cause important fish diseases***
- ***Aeromonas salmonicida*** 杀鲑气单胞菌 affects fish in marine and fresh waters



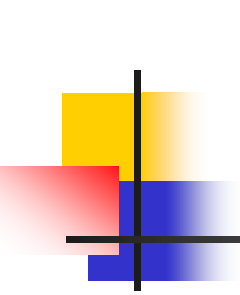
furunculosis
疔病



4. Diseases of Fish

c) Protists cause disease in fish by infections, toxins and physical effects

- **Large number of protozoan parasites infect wild, farmed and aquarium fish**
 - Often free-living or harmless parasites; conditions promoting disease largely unknown
- **Harmful algal blooms - toxins**
 - *Pfiesteria piscicida* 噬鱼费氏藻 (“cell from hell”) dinoflagellate
 - Very controversial (see MM p. 276); possible association with mass fish kills in USA as well as human mental illness
- **Non-toxic microalgae can also affect fish**
 - Large blooms reduce light penetration, decrease growth of seagrass beds where young fish grow
 - Diatoms can clog fish gills leading to death



Marine microbes have some detrimental (bad) effects:

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What can we do about them? Potentially good approaches are:

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Marine biofouling (生物污损)

Biofouling is colonization of marine surfaces by microbes with successive colonization by algae and animals

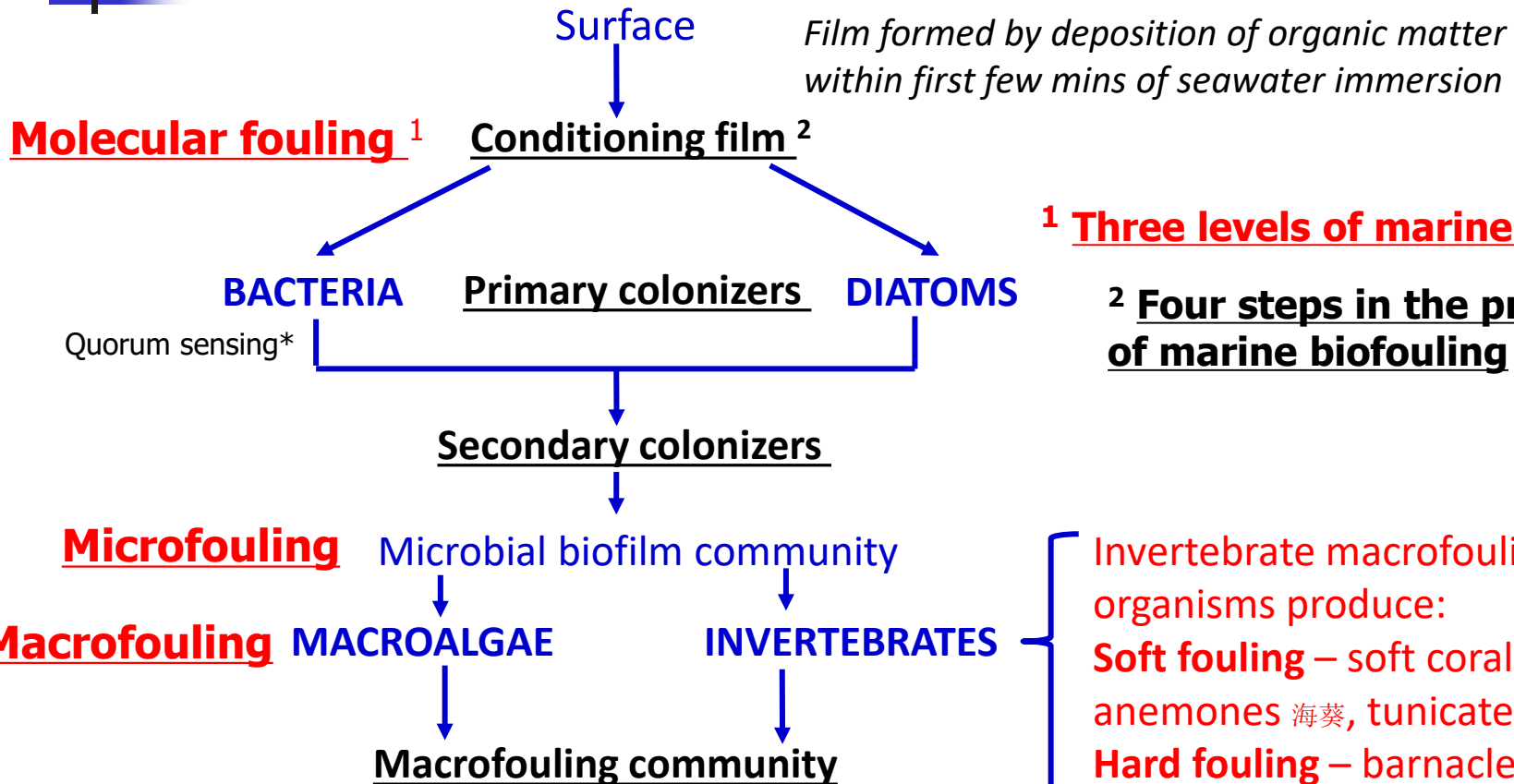
- The complex, dense community that develops leads to the biofouling of all types of marine surfaces

- Macroalgae
- Animals
- Piers and jetties 码头
- Oil-drilling rigs 钻井平台
- Fishing gear 渔具设施
- Aquaculture cages 水产养殖笼
- Boat hulls 船壳
- Coastal plants 沿海工厂

Fouling organisms ~2000 in the world;
~200 in China



Biofilm formation is often the first phase in biofouling



¹ Three levels of marine biofouling

² Four steps in the procedure of marine biofouling

Invertebrate macrofouling organisms produce:

Soft fouling – soft corals, sponges, anemones 海葵, tunicates 被囊动物

Hard fouling – barnacles 藤壶, mussels, tubeworms

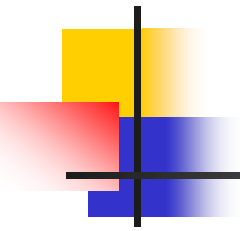
Development of biofouling, showing the critical role of microbes
(modified from MM Fig. 13.1)

Effects of biofouling



- Deterioration of materials, blockage of pipes
- Reduced efficiency of heating and cooling plants
- Interference with operation of ships and boats
 - Microbial biofilm can cause 1-2% increase in drag
 - Macroalgae can increase this to 10%, hard fouling can then lead to 30-40% drag
 - Billions of dollars world-wide due to increased fuel costs, and cost of anti-fouling treatments
- Prevention?
 - Copper or tin (锡) material-- toxic effect
 - Tributyl tin (三丁锡, TBT) effective coating for ship hulls, but banned due to environmental problems
 - Need for environmentally-friendly antifouling biotechnology
- **Positive role of biofouling:**
 - Microbial products stimulate the settlement and metamorphosis (变态) of algae and invertebrate larvae (幼虫), which is very important in marine ecology (e.g. in reef formation) and aquaculture (e.g. in settlement of mussels in suspended rope culture)





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Biodeterioration 生物腐蚀

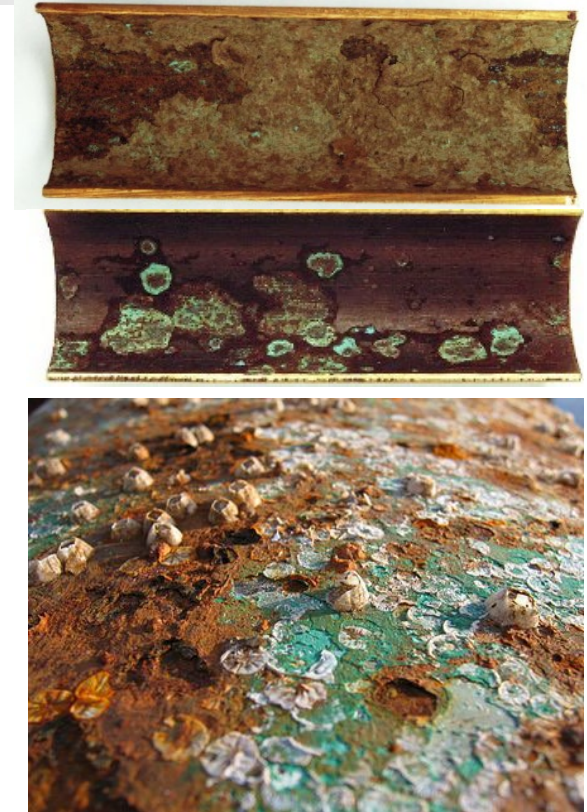
- Microbially induced corrosion occurs as the results of **the activities of microorganisms within biofilm** on metals, alloys, and composite materials
- **Sulfate-reducing bacteria (SRB)** are the main cause of marine corrosion
- Mixed consortia with **fermentative acid-producing bacteria**
 - Supply acetate as a substrate for SRB metabolism

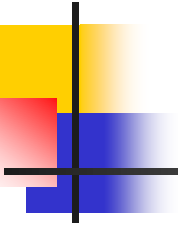
Problem for the off-shore petroleum industry –

- Pipelines and platforms corrode
- The “souring” of crude oil with H_2S

Solution-----

- Monitoring of levels of culturable acid-producing bacteria and addition of biocides (生物农药) helps to control the problem
- Testing for anaerobic SRB is more difficult because of the need for anaerobic culture; development of molecular test kits for SRB would be useful
- Biotech solution needed as large-scale use of biocides at sea is problem





Biodeterioration 生物腐蚀

Spoilage (腐败) of seafood products

- Fish and shellfish deteriorate very rapidly due to microbial activities
- Normal bacterial inhabitants of the gut and skin of fish decompose substrates in fish tissues, released by autolytic processes soon after fish death
 - *Pseudomonas* 假单胞菌, *Shewanella* 希瓦氏菌, *Alteromonas* 交替单胞菌, etc.
 - Psychrophilic strains of *Pseudomonas* and *Shewanella* become dominant when fish is chilled on ice
- Production of ammonia 氨, amines 胺, ketones 酮, sulfur 硫 compounds give rise to bad smell and flavour
- Bacteria grow rapidly to $>10^8$ CFU per gram of fish tissue
- Metabolic consortia develop in mixed microbial community
 - e.g. lactic acid bacteria (e.g. *Lactobacillus*) degrade arginine to ornithine, which is then used by enterobacteria 肠杆菌

Biodeterioration 生物腐蚀

Extending shelf-life of seafood products

- Traditional methods: salting and smoking
- Packaging in CO₂-modified atmosphere to suppress growth of respiratory bacteria
- Antibiotics were once added to ice on board fishing vessels and in markets – now prohibited due to concerns over antibiotic residues and antibiotic resistance
- Biotechnology approaches may offer new methods of control



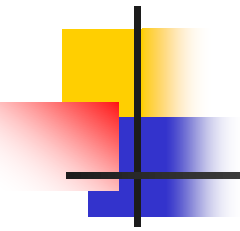
Packing herring for salting 1920s



Fish processing



Modern fish packing factory, France



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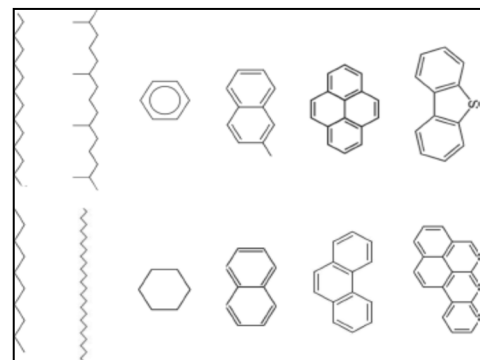
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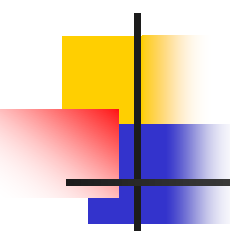
Oil and other chemical pollution



- **Oil pollution of the marine environment is a major problem**
 - 3 billion tonnes crude oil extracted p.a., ~0.1% ends up in the sea
 - Pollution by oil tanker and platform accidents is of great concern
- **A range of microbes are responsible for biodegradation of oil at sea**
 - Petroleum is a complex mixture predominantly of hydrocarbons
 - Large quantities of crude oil seep naturally into oceans so microbes in >160 genera have evolved mechanisms to degrade it
 - most hydrocarbon-degrading microbes are heterotrophs belonging to gammaproteobacteria e.g. *Pseudomonas*, *Acinetobacter* 不动杆菌
 - also some Cyanobacteria and one algal genus
 - in anoxic environments SRB and archaeal methanogens
 - Breakdown of oil depends on consortium of microbes, each transforms a particular fraction of the oil

Some petroleum compounds



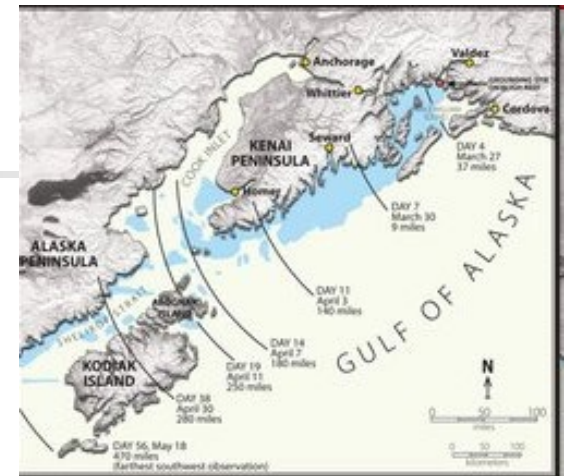


Biodegradation and Bioremediation

- **Biodegradation** is the breakdown of a complex chemical through biological processes
- **Bioremediation** is the use of biologically mediated processes to remove or degrade pollutants from specific environments
- In the case of oil pollution, the hydrocarbons provide a C source for bacteria, but oil is deficient in N and P
- This limits the rate of natural biodegradation
- The most successful approach to bioremediation has been to speed up the natural processes by addition of inorganic or organic nutrients as **fertilizers** 肥料

Exxon Valdez oil spill 1989

- Exxon Valdez oil tanker ran aground, releasing 11 million gallons* oil, contaminated 500 km coastline
- The largest oil spill clean-up ever mobilised
- One of the first demonstrations that simple addition of N and P fertilizer significantly reduced oil contamination



Spill area totalled 11000 square miles; Prince Williams Sound, Alaska



http://www.eoearth.org/article/Exxon_Valdez_oil_spill

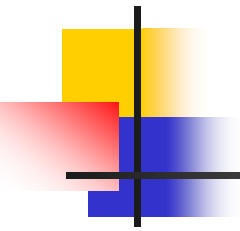
* 1 gallon=3.8 liter

Bioremediation of Exxon Valdez oil spill: N and P fertilizers added to increase the activity and number of indigenous hydrocarbon-degrading bacteria



Prior to bioremediation

Post-remediation

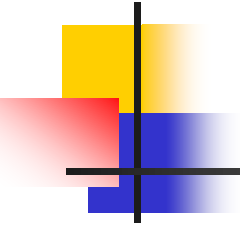


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Biotechnology

- Biotechnology is the application of science and technology to provide goods and services through use of biological agents
- Major economic benefits come from the exploitation of marine microbes, due to their enormous diversity and range of metabolism

Enzymes from marine microbes

- Enzymes are widely used by industry: market is many billion US\$
- The most successful have been isolation of enzymes from extremophiles e.g. for processes requiring high or low temperatures

Source	Product	Applications
Thermophiles	<i>Amylases</i> , lipases, proteases	Baking, brewing, food processing
	DNA polymerases	PCR
	Lipases, proteases	Detergents – stain removal
Halophiles	Bacteriorhodopsin 细菌视紫红质	Bioelectronic devices, optical switches
	Lipids	Liposomes for drug delivery, cosmetics
	Surface-layers	Ultrafiltration, electronics, polymers
Psychrophiles	Ice nucleating proteins 冰核蛋白	Artificial snow, frozen food processing
	Lipases, proteases	Cold-water detergents

Enzymes from marine microbes

- Discovery of thermostable DNA polymerases was a great scientific advance
- Huge market for PCR enzymes, kits and equipment
- *Taq* was original; alternative enzymes superior depending on application
- Enzymes from hyperthermophilic vent bacteria have greater stability and activity at higher temps

SOME THERMOSTABLE DNA POLYMERASES

DNA polymerase	Organism	Source*	Half-life at 95°C (min)
<i>Taq</i>	<i>Thermus aquaticus</i> 栖热水生菌	T (N or R)	40
<i>Pfu</i>	<i>Pyrococcus furiosus</i> 激烈热球菌	M (N)	120
Vent™	<i>Thermococcus litoralis</i> 嗜热高温球菌	M (R)	400
Deep Vent™	<i>Pyrococcus GB-D</i>	M (R)	1380

*M, marine hydrothermal vent; N, natural; R, recombinant; T, terrestrial hot spring

Pharmaceutical compounds from marine microbes

- Marine microbes are a rich source of biomedical products
- Many of our most successful drugs are metabolites from soil bacteria and fungi
- However companies moved away from natural product-based drug discovery as too expensive for low profits
- Now there is a resurgence of interest in **natural products** from marine habitats as they are likely to yield many new compounds
- So far most agents are produced by cyanobacteria and actinomycetes

Agent	Organism	Activity
Abyssomicin	<i>Verrucispora</i> sp. (A)	Broad-spectrum antibiotic
Apratoxins阿普拉毒素	<i>Lyngbya</i> sp. (C)	Antitumor
Bryostatins	<i>Candidatus</i> <i>Endobugula sertula</i> (B)	Antitumor, treatment of leukemia 白血病, Alzheimer's disease 阿尔兹海默症
Largazole	<i>Symploca</i> sp. (C)	Antiepileptic 抗癫痫, prevents mood swings
Salinosporamide	<i>Salinospora</i> sp. (A)	Antimalarial 抗疟, anticancer

Marine microbes as sources for biofuels

- Due to reduced supplies of fossil fuels & concerns about pollution, interest in using renewable biological materials
- Crops could be fermented to alcohol. However, the costs, competition for land use, and the ecological consequences limit its development.
- Microalgae are promising new sources of biodiesel fuel
 - easy to cultivate,
 - require simple nutrients and sunlight
- The lipids and fatty acids in membranes are the starting material
- The algae are lysed to extract the oil



New approaches to antifouling: quorum quenching

群体感应淬灭

- Interest in developing natural products as antifouling agents due to environmental problems of tributyl tin paints
- Products from marine organisms should
 - have low toxicity
 - be effective at low concentrations
 - break down quickly in seawater
- Hundreds of molecules investigated: best are from sessile marine organisms:
 - sponges, corals, seaweeds
- Reason is that they often have mechanisms to avoid overgrowth by other organisms: colonization on their surfaces would interfere with availability of nutrients or light



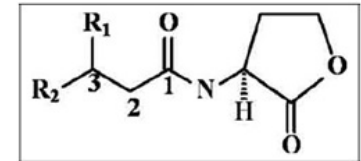
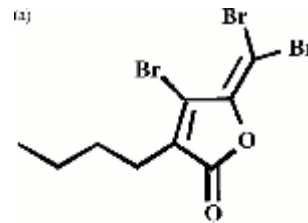
New approaches to antifouling: quorum quenching 淬灭



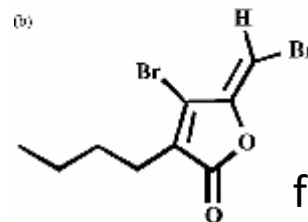
- It was noted that seaweed *Delisea pulchra* (一种海洋红藻) usually has a very clean surface without biofilm
- *Delisea pulchra* produces **brominated furanone** (溴代呋喃酮) compounds that prevent biofilm formation by interfering with quorum sensing (= “quorum quenching”)
 - furanones have structure very similar to acyl homoserine lactones AHLs
 - they prevent binding of AHLs to LuxR, so quorum sensing cannot be activated
 - this inhibits processes requiring quorum sensing e.g. biofilm formation, bioluminescence



Delisea pulchra



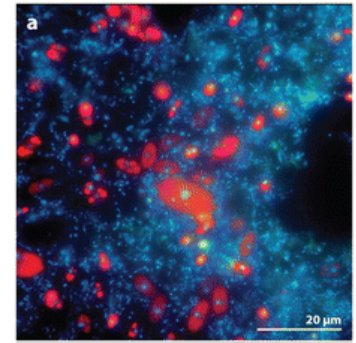
AHL



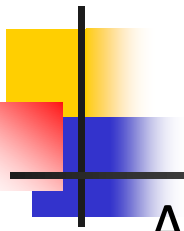
furanones

New approaches to antifouling

- Furanones have been synthesized and patented by a company
- Products where they may be used include:
 - marine antifouling paints
 - antifouling in water treatment plants
 - integration into plastics to prevent biofilm formation on contact lenses and medical implants
 - drugs to control infection in humans
- In contrast, the surfaces of some algae are colonized by *beneficial* multispecies bacterial biofilms that appear to inhibit fouling by invertebrate larvae or spores of other algae
 - antifouling coatings for plastic surfaces developed by mixing extracts of marine isolates of *Bacillus*, *Pseudomonas*, *Streptomyces* with water-based resins – “living paints”



Algal biofilm: bacteria (blue); diatoms (red)



Applications of microbial biotechnology in aquaculture

A few of the roles that the microbial biotechnology is playing in improving the health of marine animals in aquaculture (from MM Table 14.5)

Health factor	Practices
Hygiene 卫生	Bioremediation to improve water quality
Nutrition	Immune stimulants as feed additives Probiotics for growth promotion and disease prevention
Diagnosis of disease	Development of rapid methods: e.g. PCR assays, antibody methods
Eliminating vertical transmission	Testing egg and sperm for pathogen: e.g. PCR assays, antibody methods
Antimicrobial treatment	New antibiotics and synthetic agents. Phage therapy. Sensitivity testing methods
Vaccination	New vaccines: immersion, oral and injectable. DNA vaccines.
Genetic improvement of stock	Transgene-incorporation of genes for disease resistance and growth promotion



Objectives for this lecture:

At the end of this lecture, you should be able to:

Diseases (as pathogens)

- Microbial diseases of human (major human pathogens, HABS)
- Microbial diseases of marine mammals
- Microbial diseases of marine invertebrates
- Microbial diseases of fishes

Biofouling 生物污损

- Marine biofouling: process and effects

Biodeterioration 生物腐蚀

Bioremediation 生物修复

Biotechnology 生物技术

Application of marine microbes and their products including:

- Enzymes, pharmaceutical products, biofuel, antifouling agents, vaccines, etc.



期末考试题型

英文试卷，题型如下：

1. 10道判断（每个1分，共10分）
2. 10道单选（每个1分，共10分）
3. 5道多选（每个3分，共15分）
4. 5道简答（每个10分，共50分，中英文作答均可）
5. 1道实验设计题（15分，中英文作答均可）

考核范围：

秋冬全部的实验，冬学期理论课